

# What Does a Persona Do? Cognitive Trait Decomposition of Authoring Identity in AI Creative Task Generation

[Author]  
[Institution]  
USA  
[email]

## Abstract

Domain-specific persona prompts are a natural choice for AI creative generation tasks: instruct the model to adopt the identity of a practitioner, and domain conventions should follow. Yet in a controlled puzzle authoring study, a bare professional label (“*You are an artist*”) outperformed an equally sparse domain-specific label (“*You are a puzzle designer*”) by 15.6 points on a seven-dimension weighted rubric (/45), scoring 40.3 against 24.7. The cognitive mechanism behind this reversal is unknown: the artist persona encodes no puzzle design knowledge, yet produces categorically superior output. Prior work treats persona prompts as holistic units; no study has decomposed a persona into constituent cognitive traits and identified which subset is causally responsible for a quality outcome.

We address this gap through a three-phase decomposition study. Phase 1 isolates each of six candidate traits—experience-first thinking, surprise instinct, visual thinking, elegance drive, rigor and verification, and systematic thinking—as a single imperative sentence evaluated by an expert panel on the same weighted rubric. Phase 2 applies leave-one-out ablation from the full artist persona, measuring each trait’s structural contribution. Phase 3 performs cumulative reconstruction from the null baseline, adding traits in tier order to identify the minimum sufficient set. All conditions are evaluated by a three-member C8 expert panel on the same seven-dimension weighted rubric ( $\geq 33$  passing threshold).

Experience-first thinking (T1) is architecturally irreplaceable: removing it from the full artist persona causes a catastrophic  $-29.4$  point drop (from 41.7 to 12.3), producing a list of domain terms with no solver mechanism rather than a degraded puzzle. Elegance drive (T4) and rigor with verification (T5) are load-bearing: removing either drops the score below threshold ( $-11.7$  and  $-12.7$ , respectively). In Phase 1, visual thinking (T3) and surprise instinct (T2) pass in isolation (35.7 and 34.7, respectively), while elegance drive (T4, 26.0) and systematic thinking (T6, 22.3) fall at or below the domain-specific baseline—T4 because stripping without an experience-first target removes the solver’s journey along with the clutter. Phase 3 reconstruction confirms that T1 + T4 + T5 is the minimum sufficient set: their cumulative combination (R3) is the first condition to cross

the passing threshold, at 35.7/45. Visual thinking, surprise instinct, and systematic thinking are quality amplifiers whose removal the structure survives.

This paper makes three contributions. First, it provides the first trait-level decomposition of a creative persona prompt effect, replacing a holistic persona result with a causal account. Second, it empirically identifies a minimum sufficient trait set (T1 + T4 + T5)—three sentences encoding experience-first framing, elegance drive, and verified rigor—that crosses the quality threshold and generalizes to any creative generation task where audience experience is the primary design constraint. Third, it demonstrates that domain-naive artistic orientation outperforms domain-specific expertise for creative AI generation, and explains why: domain labels activate template conventions, while artistic orientation activates the experience-design stance that makes domain knowledge serve the audience rather than serve itself.

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## 1 Introduction

Two bare persona prompts. One task. Radically different outputs.

In a controlled authoring study [? ], a large language model was given the instruction “You are an artist” and asked to design a puzzle hunt feeder puzzle around the answer TOWER, with access to a locked Age of Empires II world file as its only domain resource. The result was *Damaged Inscriptions*: five stone tablets rendered in monospace, each missing one letter from an AoE2 game term, the missing letters reading in sequence to spell TOWER. The mechanism — reconstruction of a damaged inscription rather than extraction of a letter from a known word — was unlike anything else in the eight-puzzle study. The panel of three domain experts, evaluating on a seven-dimension weighted rubric (/45, pass  $\geq 33$ ), scored it 40.3/45.

The same model, given the instruction “You are a puzzle designer,” produced *The Indexed Garrison*: five AoE2 clues with explicit position markers, a visible extraction table, an acrostic reading TOWER. Clear, solvable, correct—and scored 24.7/45. The difference, 15.6 points, is larger than the gap between any two adjacent reviewer ablation conditions in the companion reviewer-profile depth study [? ].

The anomaly is structural. “Puzzle designer” carries domain knowledge that “artist” does not. The advantage should run the other way.

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## 1.1 The Question

The central question this paper addresses is not whether persona prompts affect AI creative output—that they do is now well-established [?]—but *which component of a persona is doing the work*. When a single word (“artist”) produces a 15.6-point quality advantage over a comparably sparse domain-specific label (“puzzle designer”), the gain cannot be attributed to domain knowledge, because the artist persona encodes none. It must be attributed to something else—a cognitive orientation, a set of values, an implicit theory of what good work means—that the artist identity activates and the puzzle designer identity does not.

Prior work treats personas as holistic units. A role label is applied, the model responds from that role, and the quality of the response is attributed to the persona as a whole. No study, to our knowledge, has decomposed a persona prompt into its constituent cognitive traits, isolated those traits experimentally, and identified which subset is causally responsible for the quality outcome. That is the contribution of this paper.

## 1.2 Our Approach

We decompose the “artist” persona into six cognitive traits and test each one independently, using the same puzzle authoring task as the motivating study.

The six traits were identified by asking: what does “artist” activate that “puzzle designer” does not? The answer, cross-referenced against creativity research and design theory, yielded the following: (T1) *experience-first thinking*, the disposition to design toward what the other person will feel before deciding what the puzzle will contain; (T2) *surprise instinct*, the drive to produce the moment of “wait, really?”; (T3) *visual thinking*, a preference for images and spatial relationships as the first representational format; (T4) *elegance drive*, the practice of removing anything that is not strictly necessary; (T5) *rigor and verification*, the habit of checking claims before declaring work done; and (T6) *systematic thinking*, the discipline of building a coherent structure before filling in details.

Each trait is operationalized as a single imperative sentence appended to the bare task prompt, enabling clean experimental manipulation. Three phases of experimentation test the traits in complementary ways. Phase 1 (trait isolation) runs each trait in isolation—one sentence, one agent, one puzzle—and evaluates the result against the AA and A0 controls. Phase 2 (leave-one-out) runs the full artist persona with one trait removed per condition, identifying which trait’s absence most damages quality. Phase 3 (cumulative reconstruction) starts from the null baseline and adds traits in tier order until artist-level performance is reached, identifying the minimum sufficient trait set.

## 1.3 Key Findings

The experimental results yield four findings that together constitute a causal account of the artist advantage.

First, no single trait in isolation replicates the artist’s mechanism novelty. All six trait-isolation conditions remain within the letter-indexing and acrostic mechanism family; the damaged-inscription mechanism that AA invented requires a trait combination rather than any individual trait. The highest isolation score, T3 (visual

thinking), reaches 35.7/45—substantially below AA’s 40.3 and still mechanically conventional.

Second, experience-first thinking is irreplaceable. Removing T1 from the full artist persona produces a 29.4-point drop (41.7 to 12.3), the largest catastrophic failure in the study. The result is not a degraded puzzle but an absent one: without the experience-first orientation, the model generates a list of AoE2 terms with no clue layer, no solver task, and no mechanism. No other trait removal approaches this severity. Removing elegance drive produces the next-largest drop at 11.7 points; removing rigor and verification drops 12.7 points.

Third, the minimum sufficient trait set is T1 (experience-first) + T4 (elegance drive) + T5 (rigor and verification). Phase 3 reconstruction confirms that adding these three traits in order produces the first threshold crossing at 35.7/45—a passing score no individual trait achieves alone. Tier 3 traits (visual thinking, surprise instinct, systematic thinking) each contribute 4–8 additional points of quality above the minimum sufficient set, but none is structurally irreplaceable.

Fourth, domain labels activate template-matching; artistic orientation activates experience-design. The puzzle designer persona produced an acrostic with an extraction table because that is the template the label activates. The artist persona produced a damaged-inscription mechanism because it asked first what the solver would *do*, not what structure would yield the answer. This distinction—between designing toward a structure and designing toward an experience—is the operative difference between the two personas, and it is instantiated by the experience-first trait.

## 1.4 Broader Significance

The practical implication extends well beyond puzzle hunt authoring. The three traits that constitute the minimum sufficient set—experience-first framing, elegance drive, and rigor and verification—are domain-agnostic quality dispositions. They do not encode knowledge of puzzle mechanisms, AoE2 unit costs, or letter-extraction conventions. They encode a way of working that makes domain knowledge serve the audience rather than serve itself. A prompt engineer adding T1+T4+T5 to any creative generation task—marketing copy, product design specifications, legal plain-language drafting, training material authoring—is not adding puzzle domain knowledge; they are adding the three-trait cognitive stance that the artist label activates as a bundle. The “artist” label works not because artists know puzzles, but because artistic training cultivates the exact dispositions that produce quality in any domain where a human being will eventually have to experience the output.

## 1.5 Contributions

This paper makes the following contributions.

- (1) **First trait-level decomposition of a persona prompt effect.** We decompose the artist advantage over the puzzle designer baseline into six constituent cognitive traits and measure each trait’s independent contribution to authoring quality, identifying the causal structure behind a persona prompting result that holistic analysis cannot explain.
- (2) **Empirically identified minimum sufficient trait set.** Phase 2 and Phase 3 experiments jointly confirm that T1 +

T4 + T5 constitute the minimum sufficient set for passing-threshold puzzle quality—a finding generalizable to any creative generation task where audience experience is the primary design constraint.

- (3) **Domain-naïve artistic orientation outperforms domain-specific expertise for creative generation.** The artist persona, encoding no puzzle design knowledge, outperforms the puzzle designer persona by 15.6 points. The trait decomposition explains why: domain expertise activates conventions; artistic orientation activates the disposition to transcend them.
- (4) **Three-sentence creative enhancement applicable across domains.** We provide the exact three-sentence trait injection (T1, T4, T5) that crosses the quality threshold, with empirical evidence of its effect and a theoretical account of why it generalizes.

## 1.6 Paper Organization

Section 2 situates this work relative to persona prompting literature, creative cognition research, experience-design theory, and the minimalism tradition in design practice. Section 3 describes the trait taxonomy, prompt construction, artist persona formulation, and three-phase experimental design in detail. Section 4 reports Phase 1, Phase 2, and Phase 3 results with full score tables and mechanism novelty analysis. Section 5 interprets the findings and discusses implications for prompt engineering and creative AI system design. Section 6 summarizes contributions and directions for future work.

## 2 Related Work

### 2.1 Persona Prompting Effects on AI Output

The use of persona descriptions to condition large language model behavior has a substantial empirical literature. Zhang et al. [?] demonstrate that injecting persona descriptions into dialogue systems improves response coherence and user engagement, establishing that LLMs benefit from explicit identity framing rather than generic instruction. Park et al. [?] extend this to generative agents with persistent identity representations, showing that memory-augmented personas can sustain coherent behavior across multi-turn social interactions. Dang et al. [?] apply persona conditioning to creative writing assistance, finding that distinct persona instantiations produce meaningfully different stylistic outputs from the same underlying model, with implications for diversity in co-creative systems.

A consistent limitation across this prior literature is that persona is treated as a holistic unit. A role label is applied; the model responds from that role; output quality is attributed to the persona as a whole. The implicit theoretical assumption is that any adequate persona specification will activate the relevant knowledge and values the model associates with that identity, and that the primary design variable is *which* persona to select rather than *how* to specify it. Dang et al. acknowledge that different creative personas produce different outputs but do not ask which internal component of the persona description is responsible for particular output properties [?]. Park et al. focus on persona *persistence* over time rather than on persona *composition*—the degree to which stable behavior

across turns is attributable to specific profile elements versus the identity label as a gestalt [?].

More recent work on AI reviewer personas has shown that the *richness* of a persona specification, as distinct from its mere presence, materially affects output quality. [?] demonstrate that domain-specific profiles encoding named evaluative frameworks, characteristic concerns, and analytical vocabulary produce qualitatively different review feedback from generic expert labels—not primarily in score magnitude but in the specificity, transferability, and actionability of the frameworks introduced. Crucially, however, that work operates on the *reviewer* side of creative evaluation: it asks how persona depth affects the quality of assessment, not how persona identity affects the quality of creative generation. The present paper fills the complementary gap: we ask which components of a creator persona are responsible for creative output quality, using the same domain (puzzle hunt design) and the same evaluation infrastructure.

To our knowledge, no prior work has decomposed a persona prompt into constituent cognitive traits, isolated those traits experimentally, and identified which subset causally drives observed quality outcomes. The present paper introduces this paradigm.

### 2.2 Creative Cognition and Expertise

The relationship between domain expertise and creative output is contested in creativity research, and the artist-versus-puzzle-designer finding lands directly in the middle of that debate.

Csikszentmihalyi's systems model [?] situates creative contribution at the intersection of three entities: the individual (person), the symbolic domain (accumulated knowledge and conventions), and the social field (gatekeepers and evaluators who recognize genuine novelty). On this account, domain expertise is necessary but not sufficient for creativity—the individual must internalize the domain's conventions deeply enough to transform them, rather than merely reproduce them. The artist persona finding is consistent with this model: the puzzle designer who produces an acrostic extraction table has internalized the domain's conventions but has not transformed them. The artist, lacking those conventions, designs from first principles and produces a structurally novel mechanism.

Weisberg's work on creative expertise [?] complicates this picture. Weisberg argues against the romantic view of creativity as transcendent novelty, proposing instead that creative advances are produced by deep domain knowledge combined with incremental variation. On this account, the artist's advantage should be impossible: without puzzle domain knowledge, the artist cannot produce meaningful novelty. That the artist nonetheless produces the study's only novel mechanism suggests that Weisberg's account, while broadly accurate for expert practitioners, underweights the contribution of *design stance*—the cognitive orientation from which the creator approaches the task—relative to domain content. The trait decomposition reported here provides a mechanistic account of what "design stance" consists of and which of its components are load-bearing.

Sawyer's work on improvisation and creative constraint [?] is relevant to understanding why artistic orientation may outperform domain expertise for generative tasks. Sawyer argues that experienced improvisers work within structures they have internalized

so deeply that the structures become invisible, freeing cognitive resources for experience-oriented decisions (“what should happen next for the audience?”) rather than rule-compliance decisions (“what is allowed next given the grammar?”). The artist persona activates this same reorientation: the trait-isolation results show that experience-first thinking (T1) is the primary driver of quality, while systematic thinking (T6)—the trait most analogous to rule-grammar compliance—produces the lowest isolation score in the study (22.3/45, below the puzzle designer baseline of 24.7/45). Cognitive freedom from convention, in this account, is not ignorance but a different allocation of attention.

## 2.3 Experience-Design and Human-Centered Creation

The experience-first trait (T1)—“you always think first about how the other person will experience what you make”—is the operationalization of a principle with a rich theoretical basis in human-centered design. Norman’s foundational work [?] establishes that the user’s experience of an artifact is not a downstream property of good engineering but its primary constraint: design that begins with structure and adds usability as an afterthought produces categorically inferior results compared to design that begins with the user’s mental model and builds structure to serve it. Hassenzuhl’s theory of experience design [?] extends this principle from usability to hedonic quality, arguing that the most important design question is not “what does this product do?” but “what does it feel like to use it?”—a reframing that shifts the designer’s primary object of attention from feature specification to experiential arc.

In the puzzle design literature, this principle is named explicitly by Thomas Snyder, whose review frameworks appear throughout the C8 panel evaluation data. Snyder’s central construction criterion—“would you want to solve this?”—is experience-first design applied to puzzle authoring [?]. A constructor who asks “what mechanism yields TOWER?” designs differently from one who asks “what would it feel like to discover that the answer is TOWER?” The damaged-inscription mechanism in AA is the product of the second question: the mechanism is the experience of recognition—the solver sees a gap in a word, reconstructs the missing letter from domain knowledge, and experiences a moment of identification that is simultaneously archaeological and self-confirming. This is not a mechanism overlaid on an experience; it is a mechanism that is the experience.

The leave-one-out results confirm that experience-first thinking is not merely one quality-amplifying factor among six—it is the architectural foundation of the puzzle as a solvable artifact. Without T1, the resulting “puzzle” (L-T1, scored 12.3/45) is a list of AoE2 terms with no clue layer, no mechanism, and no solver task. The model without the experience-first orientation does not produce a worse puzzle; it produces an object that is not a puzzle at all.

## 2.4 Minimalism and Reduction in Design Practice

The elegance drive trait (T4)—“you remove anything that isn’t necessary; you value spare, clean form above all”—is the operationalization of a design principle with as strong a theoretical pedigree as experience-first thinking. Dieter Rams articulated this principle as

the tenth of his ten principles for good design: “Good design is as little design as possible” [?]. In Rams’s framework, reduction is not aesthetic preference but epistemic discipline: excess material obscures the relationship between form and function, making it harder for the user to understand what the artifact does and how to use it. Antoine de Saint-Exupéry’s often-cited formulation—“Perfection is achieved not when there is nothing more to add, but when there is nothing left to take away”—makes the same point as a phenomenological principle: the experience of a well-designed artifact is the experience of inevitability, and inevitability is impossible in the presence of excess.

The leave-one-out result for elegance drive (L-T4, drop of −11.7 points, below-threshold at 30.0/45) provides empirical support for these design principles in a novel experimental context. The degraded condition does not merely score lower—it self-sabotages in a specific and diagnostic way. The closing note in L-T4 telegraphs the answer TOWER before the solver has had the opportunity to find it, converting the extraction from a discovery into a confirmation of what the solver was already told to expect. The unedited puzzle undermines its own mechanism. This is precisely the failure mode Rams predicts: excess material does not merely add noise; it corrupts the signal relationship between the designer’s intention and the solver’s experience.

Experimental aesthetics literature provides complementary evidence. Reber et al. [?] demonstrate that perceptual fluency—the ease with which a stimulus can be processed—produces aesthetic pleasure, and that sparse form generally produces higher fluency than cluttered form. The elegant puzzle is not merely more pleasing; it is more legible, and that legibility is itself a quality of the solver’s experience. The elegance drive trait operationalizes this finding as an authoring constraint: before calling work done, remove anything that reduces the clarity of the signal relationship between clue and answer.

## 2.5 Prior Work on Minimal Sufficient Conditions in Creative Systems

The Phase 3 reconstruction experiment—starting from a null baseline and adding traits until the quality threshold is crossed—is an application of the *minimum sufficient set* paradigm to creative generation prompting. This paradigm has precedents in adjacent literatures. In automated theorem proving, minimal proof certificate identification has been studied as a problem of identifying which axioms and inference steps are load-bearing and which are redundant [?]. In feature selection for machine learning, ablation-based subset identification is standard methodology. In human-computer interaction, the minimum viable prototype paradigm applies a related logic to product development [?].

What is novel in the present application is the identification of a minimum sufficient cognitive trait set for an open-ended creative generation task, using a human expert panel as the quality oracle. This differs from minimum viable prototype work in that the quality criterion is evaluative (expert panel score) rather than functional (does the feature work), and from machine learning ablation in that the “features” being tested are cognitive orientations rather than measurable input variables. The result—that T1 + T4 + T5 constitute the minimum sufficient set for passing-threshold puzzle

authoring quality, with each Tier 3 trait contributing 4–8 additional quality points—is, to our knowledge, the first empirically identified minimum sufficient trait set for a persona prompting effect in any creative domain.

The gap this work fills is therefore threefold: no prior study decomposes professional identity prompts into constituent cognitive traits, no prior study identifies the minimum sufficient trait subset empirically, and no prior study demonstrates that domain-naïve artistic orientation outperforms domain-specific expertise labels for creative generation tasks.

## 3 Methodology

### 3.1 Motivating Persona Study

The experiments reported in this paper originate in a finding from the companion study on authoring profile taxonomy [? ], which evaluated seven bare persona conditions on a controlled puzzle authoring task. Each condition received the instruction “You are a [role]” and was given the same task: design an Age of Empires II feeder puzzle whose final answer is the word TOWER, using only a locked world data file as domain reference. No puzzle design conventions, extraction templates, or hunt design principles were provided. The seven conditions were: null (AX, no role label), puzzle designer (A0), game designer (AG), programmer, doctor, artist (AA), and no job title.

The C8 panel evaluated all seven outputs on the seven-dimension weighted rubric (/45, Clarity  $\times 1$  + Solvability  $\times 1$  + Elegance  $\times 2$  + Reading Reward  $\times 2$  + Fun  $\times 1$  + Confirmation  $\times 1$  + Riven Standard  $\times 1$ , pass  $\geq 33$ ). Scores ranged from 16.3 (AX-null) to 40.3 (AA-artist), with the game designer condition (AG) second at 32.3 and all other conditions below 33. The artist was the only condition to achieve a passing score and the only condition to produce a mechanism outside the letter-indexing and first-letter acrostic family. All six other conditions, regardless of professional label, extracted TOWER via first-letter or positional indexing. The 15.6-point gap between AA (40.3) and A0 (24.7) is the motivating anomaly this paper explains.

### 3.2 Trait Taxonomy

The six cognitive traits were identified through a two-stage process. In the first stage, we conducted an introspective analysis of the artist persona: given that “You are an artist” contains no puzzle domain knowledge, what disposition or orientation could it activate that would drive a model toward experience-first mechanism design rather than template-extraction? The answer we formulated was that artistic training, as documented across design philosophy, creativity research, and practitioner writing, cultivates a small number of domain-agnostic cognitive habits: a primary orientation toward the audience’s experience, a drive to produce surprise and recognition rather than mere correctness, a preference for visual and spatial thinking as the first representational step, a discipline of removing excess until the core is exposed, a habit of verification before completion, and a practice of structural coherence before surface elaboration.

In the second stage, each candidate disposition was cross-referenced against the creativity and design theory literature. Experience-first thinking is the operationalization of Norman’s user-centered design principle [? ] and Hassenzahl’s experience-design framework [?

]. Surprise instinct connects to Berlyne’s theoretical account of aesthetic curiosity and surprise as primary drivers of aesthetic engagement [? ]. Visual thinking is documented in Arnheim’s research on visual thinking as a primary cognitive mode [? ] and in Shepard and Metzler’s foundational mental rotation work [? ]. Elegance drive is the operationalization of Rams [? ] and Saint-Exupéry’s reduction principle. Rigor and verification connects to Snyder’s construction discipline [? ]; no puzzle element should be asserted without confirming that it is factually accurate and that the extraction is uniquely determined. Systematic thinking connects to well-documented expertise advantages in structured problem decomposition [? ].

Each trait was then formulated as a single imperative sentence designed to activate the disposition as a behavioral directive without introducing puzzle domain content. The imperative formulation is deliberate: it specifies a way of working rather than a domain of knowledge, ensuring that any quality gain observed in the trait-isolation conditions is attributable to the cognitive orientation rather than to domain expertise injected via the trait sentence.

The six resulting trait prompts are as follows. T1 (Experience-First): “You always think first about how the other person will experience what you make.” T2 (Surprise Instinct): “You care deeply about surprising people—you want them to pause and think ‘wait, really?’” T3 (Visual Thinking): “You think in images and spatial relationships before you think in words.” T4 (Elegance Drive): “You remove anything that isn’t necessary—you value spare, clean form above all.” T5 (Rigor/Verification): “You check your work carefully before calling it done—you never assume it works.” T6 (Systematic Thinking): “You approach problems by breaking them into a logical structure before building anything.”

The six traits have a partial mapping onto Boden’s three creativity types [? ]: T1 (experience-first) and T4 (elegance drive) function as *exploratory* constraints, establishing the conceptual space within which the mechanism is sought; T2 (surprise instinct) and T3 (visual thinking) function as *combinatorial* operators, generating novel pairings within that space; T5 (rigor/verification) functions as *transformational* pruning, eliminating candidates that violate quality gates. T6 (systematic thinking) is structural scaffolding rather than a creativity type per se. This mapping is approximate and the taxonomy makes no claim to exhaustiveness — other creative orientations may introduce additional traits not represented here.

### 3.3 Full Artist Persona

For Phase 2 (leave-one-out) and as the reference control for all three phases, a full artist persona was constructed by integrating all six trait dispositions into a coherent character description. The integration was designed to avoid a list structure that would make individual traits visible and removable to the model; instead, the traits are woven into a continuous characterization that reads as a natural identity description. The full artist persona used across all leave-one-out conditions is:

You are an artist. You always think first about how the other person will experience what you make. You care deeply about surprising people. You think in images and spatial relationships before you think in words. You remove anything that isn’t necessary. You check

your work carefully before calling it done. You approach problems by building a coherent structure.

This is the prompt that produced the AA condition (DAMAGED INSCRIPTIONS, 41.7/45 on Phase 2 re-evaluation; published score 40.3/45 from the motivating study, within expected reviewer variance across evaluation sessions). Each leave-one-out condition removes one sentence from this paragraph, replacing the missing trait with no substitute text. The removal is exact: the remaining persona is a strict subset of the full text, preserving all other trait descriptions unchanged.

### 3.4 Phase 1: Trait Isolation

Phase 1 tests each of the six traits in isolation: one trait sentence, one agent, one puzzle authoring task. Each agent receives the bare task (“Design an Age of Empires II feeder puzzle whose answer is TOWER”) plus the single-sentence trait injection and the locked AoE2 world data file. No persona identity label (“artist,” “designer,” or otherwise) is provided—the trait sentence is the complete persona specification. Conditions T1 through T6 are thus the minimal possible operationalizations of each trait: if any quality gain is observed, it is attributable to the single sentence and nothing else.

Six agents were run in the same session as the AA and A0 controls from the motivating study, providing a shared calibration baseline. All eight puzzles (T1–T6, AA, A0) were read by the C8 panel before any scoring, with an explicit calibration note establishing the mental rank ordering across the full set before any numerical scores were assigned. This panel-reads-all-first protocol is the same calibration procedure used in the companion evaluation papers [?] and ensures that individual puzzle scores are assigned relative to the full distribution rather than evaluated in isolation.

Phase 1 addresses two specific questions. First, which single trait has the greatest independent impact on puzzle quality? Second—and more diagnostic—does any single trait replicate the artist’s mechanism novelty? If a single trait (particularly T1 or T3) independently produces a damaged-inscription-type mechanism, the mechanism novelty finding from the motivating study is attributable to a single cognitive orientation rather than to the full artist persona. If no single trait replicates mechanism novelty, the finding is that AA-level performance requires a trait combination.

### 3.5 Phase 2: Leave-One-Out

Phase 2 tests the full artist persona with one trait removed per condition. Six agents run the full paragraph minus one sentence each, producing conditions L-T1 through L-T6. The same task and world data are provided; evaluation uses the same C8 panel and /45 rubric. Because the full persona is the starting point rather than the bare task, Phase 2 measures the *marginal contribution* of each trait to a near-complete creative specification: not what each trait adds from nothing, but what each trait’s removal costs.

The leave-one-out design is the standard ablation methodology for identifying load-bearing components in a system where components interact. A trait that produces a large drop when removed is structurally irreplaceable—no other trait or combination of remaining traits compensates for its absence. A trait that produces a small drop when removed is quality-amplifying but substitutable—other

traits in the system are sufficient to maintain passing quality even without it.

Phase 2 also answers a diagnostic question that Phase 1 cannot: whether a trait that scores poorly in isolation is in fact load-bearing when embedded in the full artist context. T5 (rigor and verification) scores 27.0/45 in isolation—below passing—but its removal from the full artist persona produces a 12.7-point drop, crossing below threshold. This pattern, shared by T4 (isolation 26.0, removal −11.7), indicates that T4 and T5 are not primarily independent quality drivers; they are constraints that prevent the rest of the persona from self-sabotaging. T5 in isolation produces careful but experience-free puzzles; T5 in the full artist context prevents factual errors that would break solvability for the domain-expert panel members who constitute the C8 reviewer population.

### 3.6 Phase 3: Cumulative Reconstruction

Phase 3 starts from the null baseline (AX, no persona, no trait sentences) and adds traits one at a time in tier order as established by Phase 2 results. The reconstruction sequence is:  $AX \rightarrow +T1 \rightarrow +T4 \rightarrow +T5 \rightarrow +T2 \rightarrow +T3 \rightarrow +T6$ . Conditions are designated R0 (AX, baseline) through R6 (all six traits, equivalent to the full artist persona). R0 and R6 are in hand from the motivating study and Phase 2 re-evaluation respectively; R1 through R5 are new agents run under the same task and world data conditions.

The build order follows the tier classification implied by Phase 2 leave-one-out results: Tier 1 (irreplaceable, removal > −20 points) is added first; Tier 2 (load-bearing, removal −11 to −13 points) is added second and third; Tier 3 (quality-amplifying but substitutable, removal −4 to −8 points) is added in order of estimated Phase 2 contribution. This sequencing ensures that the reconstruction trajectory follows the steepest quality gradient—each step adds the trait whose absence was most costly at that tier—and that the first threshold crossing (quality  $\geq 33/45$ ) is identified at the earliest possible reconstruction step.

Phase 3 identifies the minimum sufficient trait set as the smallest R-step set at which quality first crosses the 33/45 pass threshold. All seven conditions (R0 through R6) are evaluated in a single C8 panel session with full calibration, preventing between-session reviewer drift from confounding the reconstruction trajectory. The Phase 3 session result thus provides the cleanest available estimate of how each tier of trait accumulation affects quality, from the floor (AX, 16.3/45 in the motivating study) to the ceiling (AA-equivalent at  $\approx 41/45$ ).

### 3.7 Evaluation Infrastructure

All experiments use the same evaluation infrastructure as the motivating persona study and the companion reviewer-profile depth paper [?]: the C8 panel (three domain-expert reviewer profiles, Dan Katz, Thomas Snyder, and Dana Young, instantiated from 80–120 line structured markdown profiles encoding their characteristic evaluative frameworks and review vocabulary), the seven-dimension weighted rubric (/45 total, pass  $\geq 33$ ), and the all-puzzles-before-any-scores calibration protocol. The same AoE2 world data file used in the motivating study provides domain reference for all authoring agents, ensuring that world-knowledge access is held constant across all conditions.

Scores are reported as three-reviewer panel averages, with individual reviewer scores reported in full in the results section for conditions where reviewer spread exceeds two points on any dimension. Because all reported scores represent single evaluation runs at temperature 1.0, scores are point estimates without sampling variance. The three-reviewer design provides a partial proxy for evaluative uncertainty: spread within the panel at any given score dimension signals genuine disagreement that averaging would suppress. The evaluation rubric's double-weighting of Elegance and Reading Reward reflects the finding from [?] that these two dimensions are the strongest tier discriminators in the hunt quality distribution, and that equal-weight rubrics systematically miscalibrate for creative work that deliberately trades Clarity and Solvability ceiling for Elegance and Reading Reward—precisely the trade-off that distinguishes AA from A0.

A methodological note on evaluation validity: the C8 panel is itself AI-instantiated—three AI reviewers operating under domain-expert profiles. Using AI judgment to assess AI creative output introduces an epistemological circularity that this study cannot resolve internally. The companion reviewer ablation study [?] provides the closest available proxy for panel validity (AI panel scores correlated with quality tier separations on 15 real puzzle hunt puzzles), but direct human expert validation of trait-decomposition quality claims remains a required future step.

## 4 Results

Three phases of experimentation converge on a single structural claim: experience-first thinking (T1) is the architectural foundation of puzzle quality, elegance drive (T4) and rigor/verification (T5) are load-bearing walls, and the remaining three traits (T2, T3, T6) are quality amplifiers whose removal the structure survives. We report the phases in sequence.

### 4.1 Phase 1: Trait Isolation

Each of the six traits was tested in strict isolation—one imperative sentence appended to the null prompt—and evaluated by the three-member C8 panel on the weighted seven-dimension rubric (/45, pass  $\geq 33$ ). The two control conditions, AA (full artist persona, the motivating study's top scorer) and A0 (null baseline), anchor the distribution. Table 1 reports all eight conditions.

**Table 1: Phase 1 trait isolation scores (C8 panel, /45). Pass threshold:  $\geq 33$ .**

| Condition | Trait               | Avg/45 | Pass           |
|-----------|---------------------|--------|----------------|
| AA        | Control (artist)    | 41.0   | Y              |
| T3        | Visual Thinking     | 35.7   | Y              |
| T2        | Surprise Instinct   | 34.7   | Y              |
| T1        | Experience-First    | 31.3   | N (borderline) |
| T5        | Rigor/Verification  | 27.0   | N              |
| T4        | Elegance Drive      | 26.0   | N              |
| A0        | Control (baseline)  | 24.0   | N              |
| T6        | Systematic Thinking | 22.3   | N              |

Two trait-isolation conditions pass the panel: T3 (visual thinking, 35.7) and T2 (surprise instinct, 34.7). Both remain 5–6 points

below AA's 41.0, a gap of roughly one full weighted dimension, and both remain within the letter-extraction mechanism family; neither produces a novel mechanism. The passing conditions confirm that structural embellishment—visual chrome on top of indexing, or ironic framing that withholds the answer until the extraction reveals it—can carry a puzzle above threshold when that embellishment is well-executed. They do not confirm that any single trait can replicate what the full artist persona produces.

Four trait conditions fail: T5 (27.0), T4 (26.0), A0 (24.0), and T6 (22.3). The ordering is instructive. T5 (rigor/verification) and T4 (elegance drive) both fall just above or at the baseline level, suggesting that in isolation each trait shapes the puzzle's surface without generating the experiential orientation required for quality. T4's failure is particularly diagnostic: the stripped-bracket presentation eliminated discovery entirely. The puzzle presents five clues with the answer letters visible in brackets before the solver has identified a single game term, rendering the extraction superfluous. Stripping to minimum without experience-first orientation strips the solver's journey along with the clutter. The elegance collapses into transparency.

T6 (systematic thinking, 22.3) scores the lowest of all eight conditions, falling below the null baseline. The procedure-as-puzzle format—a five-row table specifying clue, identification step, position parameter, and letter output—is precisely the format that performs worst on the rubric's highest-weighted dimensions, Elegance ( $\times 2$ ) and Reading Reward ( $\times 2$ ). Panel reviewer Snyder applied the Computer Test directly: executing a lookup table is not a deductive act, and a puzzle that can be solved by mechanical execution rather than knowledge and inference fails to meet the construction standard for the reviewed panel. The systematic trait, tested alone, produces the most procedurally explicit puzzle in the study, and procedure is the opposite of what the Elegance and Riven Standard dimensions reward.

A mechanism analysis across all eight conditions reveals a finding with direct implications for Phase 2 and Phase 3. All six trait-isolation conditions remain within the letter-extraction family. T3 (visual thinking) achieves the farthest escape from bare indexing in surface form, with a diagram header and circled positional indices that create a different experience of the same underlying mechanism. T2 (surprise instinct) achieves the deepest semantic escape: the solver spends the puzzle naming monks, archers, and cavalry units—none of which are towers—and extracts TOWER as a structural punch line. The extraction is still positional indexing, but the experiential relationship between clue and answer creates genuine surprise. Yet the underlying cognitive action in both T3 and T2 remains syntactic: identify the AoE2 term and count to letter N.

The artist condition (AA) is the only puzzle in the study whose underlying cognitive action is categorically different. The solver is asked not “which letter is at position N in this word?” but “what word is this damaged inscription a version of?” The first question is syntactic extraction; the second is semantic reconstruction. This distinction—restoration of partial information versus indexing into complete information—cannot be derived from any single trait injection. No trait alone caused the artist to design a mechanism requiring semantic reconstruction rather than syntactic extraction. The minimum sufficient mechanism requires a combination of traits, which Phases 2 and 3 identify.

## 4.2 Phase 2: Leave-One-Out

Seven conditions were evaluated in Phase 2: the full artist persona (AA, reference score 41.7) and six versions of AA with one trait removed per condition. The reference score of 41.7 reflects slight reviewer variance from the Phase 1 reading (41.0); the leave-one-out panel re-scored AA fresh after calibrating against all seven Phase 2 puzzles. Table 2 reports all conditions sorted by drop magnitude.

**Table 2: Phase 2 leave-one-out scores. Drop computed against AA reference (41.7). Pass threshold:  $\geq 33$ .**

| Condition | Missing Trait       | Avg/45 | Drop  | Pass       |
|-----------|---------------------|--------|-------|------------|
| AA        | None (reference)    | 41.7   | —     | Y          |
| L-T3      | Visual Thinking     | 37.3   | −4.4  | Y          |
| L-T6      | Systematic Thinking | 33.7   | −8.0  | Y (barely) |
| L-T2      | Surprise Instinct   | 33.3   | −8.4  | Y (barely) |
| L-T4      | Elegance Drive      | 30.0   | −11.7 | N          |
| L-T5      | Rigor/Verification  | 29.0   | −12.7 | N          |
| L-T1      | Experience-First    | 12.3   | −29.4 | N          |

The L-T1 result is not a quality degradation—it is a categorical failure. Removing experience-first orientation from the full artist persona produces a drop of 29.4 points, more than double the next-largest drop (L-T5, −12.7). The solver-visible content of the resulting puzzle is five AoE2 terms in a column and a single question: “What word do they spell?” No clues are present. No mechanism is declared. The model, operating with all other traits active—visual thinking, surprise, elegance, rigor, systematic structure—but without the fundamental orientation toward the solver’s experience, produces a list of answers rather than a puzzle. Reviewer Katz called this a contract violation; reviewer Snyder applied the Computer Test immediately and confirmed that trivial string processing solves the “puzzle” completely; reviewer Young observed that there is no moment where the world resolves because there is no solver journey to resolve. The artist label, stripped of its experience-first core, produces concept art with no solver path. “You are an artist” without experience-first orientation does not produce puzzles; it produces taxonomies.

L-T4 (−11.7, 30.0/45) and L-T5 (−12.7, 29.0/45) both fall below threshold. Each failure has a specific and diagnostic shape. The L-T4 puzzle is functional but unedited: a framing paragraph, an italic subtitle, and a closing note that telegraphs the answer before the solver finds it are all present. The closing note is the critical failure—it converts a potential discovery moment into a confirmation of what the solver has already been told to expect. Elegance drive, when absent, does not merely reduce polish; it enables the constructor to self-sabotage by including material that actively undermines the extraction. Every other trait is present, and they cannot compensate. The L-T5 puzzle fails through two specific unverified facts: the Watch Tower’s Feudal Age advancement trigger and the in-game label “Stone Wall” were flagged by the panel as potentially incorrect. Reviewer Snyder treated a factual error in a clue as a structural failure: if the clue is wrong, the entry has no valid unique answer, and the puzzle is unsolvable for an informed solver. Reviewer Katz applied the Blame-the-Player test and found it inverted—the solver with domain expertise will blame the puzzle, not themselves, which is the correct inversion of the desired relationship.

L-T3 (−4.4, 37.3) sustains a comfortable pass. The scout-dispatch narrative in the L-T3 puzzle provides a partial substitute for the spatial discovery that visual cueing would deliver. Five battle reports read as fiction until the solver recognizes the structural trick, generating misdirection through prose rather than layout. Reviewer Young’s objection—that the mechanism is given at the bottom rather than found spatially—is real but not fatal; the narrative misdirection replaces one layer of discovery with another. Language can approximate spatial guidance.

L-T6 (−8.0, 33.7) and L-T2 (−8.4, 33.3) both barely pass. L-T6’s deficit is structural: without systematic thinking, two entries permit multiple valid resolutions, introducing binding-choice ambiguity at entries 4 and 5. The puzzle passes only because the atmospheric prose quality (a distinct contribution of the remaining traits) is high enough to carry the panel above threshold despite construction failures that Snyder identified as uniqueness violations. L-T2’s pass is similarly narrow: the mechanism is declared before the first clue rather than discovered through solving, reducing the extraction from a revelation to a confirmed task completion. The Tier 3 traits are real contributors to quality; they are simply not structurally irreplaceable in the way T1, T4, and T5 are.

The three-tier structure is confirmed by Phase 2 alone: T1 is irreplaceable (removal = −29.4, categorical failure), T4 and T5 are load-bearing (removal drops below threshold, −11.7 and −12.7 respectively), and T2, T3, T6 are quality-amplifying but substitutable (removal maintains a pass, −4.4 to −8.4).

## 4.3 Phase 3: Reconstruction Curve

Phase 3 begins from the null baseline and adds traits in tier order—T1 first, then T4, then T5, then the three Tier 3 traits—evaluating the result at each step. The full artist persona (AA, 42.3 in Phase 3 calibration) serves as the ceiling. Table 3 reports the full reconstruction curve.

**Table 3: Phase 3 reconstruction curve. Traits are accumulated in tier order. First threshold crossing at R3.**

| Code | Active Traits      | Avg/45 | Pass | Notes             |
|------|--------------------|--------|------|-------------------|
| R0   | none               | 22.7   | N    | null baseline     |
| R1   | T1                 | 28.3   | N    |                   |
| R2   | T1, T4             | 30.3   | N    |                   |
| R3   | T1, T4, T5         | 35.7   | Y    | first pass        |
| R4   | T1, T4, T5, T2     | 37.3   | Y    |                   |
| R5   | T1, T4, T5, T2, T3 | 40.3   | Y    |                   |
| R6   | full artist (AA)   | 42.3   | Y    | reference ceiling |

The curve rises in two distinct rhythms. The first three steps (R0 to R3) are structural: each trait added transforms what kind of artifact the puzzle is. The final three steps (R4 to R6) are qualitative: each trait added raises the score but does not change the fundamental category of the artifact.

R0 to R1 (+5.6 points): T1 alone produces a real improvement over the null baseline. The puzzle “FIELD DISPATCHES” has a live voice—each clue is a field report from ongoing combat, deploying a unit in a moment of tactical action rather than querying a database. The mechanism (take the first letter of the second word) is withheld

and consistent, and the Woad Raider and War Elephant entries in particular place the solver inside the game rather than outside it. The improvement over R0 is genuine: this is an authored puzzle, not an assembled one. It does not pass, but it is recognizably a puzzle.

R1 to R2 (+2.0 points): T4 reshapes the container. The mechanism is stripped from its declaration and withheld; the column-indexing scheme is now discovered rather than given. The Town Center entry, with its central-building status and the letter W hidden at position three, is an authored decision the constructor chose rather than a position the procedure dictated. The improvement is structural economy: the puzzle is more legible because there is less in the way between clue and mechanism. It does not yet pass because without T5, the numbers are decorative rather than functional.

R2 to R3 (+5.4 points, first threshold crossing): T5's contribution is decisive. In CALIBER, the numbers embedded in the clues—the one-Monk rest rule, the second upgrade tier, the elite archer's maximum range, the two-point team bonuses—are the game's actual values. They simultaneously identify the answer and perform the extraction. The numbers are not invented to force a letter; they are the game's own precision repurposed as mechanism. Reviewer Snyder applied the Computer Test and found it impossible to pass: no script could generate CALIBER's structure, because its structure is the game's own factual architecture. The title lands cleanly (CALIBER = measurement precision = index precision), and the symmetric clue structure at entries four and five (same syntactic frame, deliberately different answers) rewards knowledge while penalizing guessing. This is the minimum sufficient set: T1 (a live solver inside a live game), T4 (a stripped container that withholds until the solver discovers), T5 (facts that are the mechanism, not decoration). The threshold is crossed at 35.7/45.

R3 to R4 (+1.6 points): T2 adds structural surprise. In GAPS, the answer TOWER appears in plain text in the very first clue ("WA\_CH TOWER") before the solver begins, invisible because the solver is looking for a missing letter, not a hidden answer. The device exploits the solver's own attention allocation rather than deceiving through misdirection. It is a genuine surprise, built into the structure rather than imported as flavor.

R4 to R5 (+3.0 points): T3 makes the mechanism spatially visible. The shaded column in ASSAULT is the mechanism rendered as a spatial object—the solver sees the structural claim before reading a word. Reviewer Young gives ASSAULT the highest reading-reward marks of any reconstruction condition because the layout is the argument; the visual grammar does not illustrate the puzzle, it constitutes it.

R5 to R6 (+2.0 points): the full artist persona ties the five entries into a coherent argument. The content of DAMAGED INSCRIPTIONS spans the full range of the game (siege engine, archer, raider, warrior, worker), and the closing sentence names that span explicitly, making TOWER feel inevitable rather than arbitrary. Every element is load-bearing—remove any tablet and the extraction fails—and the mechanism (restoration of a damaged inscription) is structurally identical to the theme (time and weather wore away exactly one letter from each inscription). The puzzle does not have a mechanism applied to a theme; it has a mechanism that is the theme.

The minimum sufficient set is confirmed: T1 + T4 + T5, crossing the threshold at R3 (35.7/45). The Tier 3 traits contribute 4.6

additional points in aggregate (R3 to R5, 35.7 to 40.3), and the full artist persona adds a further 2.0 points (R5 to R6), reaching the study's ceiling at 42.3/45. No single Tier 3 trait is irreplaceable; all three together account for approximately half the gap between the minimum sufficient set and the artist ceiling.

## 5 Discussion

The three experimental phases converge on a finding that is simultaneously specific—it concerns puzzle hunt authoring on a 45-point weighted rubric—and general: domain labels tell an AI what kind of output to produce, while artistic orientation labels tell it to ask what the person on the other end will feel before deciding on output format at all. We discuss the implications of this distinction and its structural supports.

### 5.1 Domain Labels Versus Artistic Orientation

The central finding of this study is a contrast between two cognitive stances activated by two single-word persona labels. "Puzzle designer" activates a domain template: conventions, formats, mechanism types, and quality criteria accumulated by practitioners of the field. "Artist" activates an experience-design orientation: the disposition to ask "what does this person do, and how does it feel?" before asking "what does this artifact contain?"

This is not a puzzle-specific finding. A domain label tells the AI what kind of output to produce—the format, the genre, the conventions that practitioners of the domain would recognize as correct. An artistic label tells the AI to make experience-design the primary constraint, before format, genre, and convention are decided. For creative tasks, the second question is more fundamental, because format and genre conventions are means rather than ends. They exist to serve the experience of the person who will encounter the artifact. When they are activated as ends in themselves—which is what a domain label does—they constrain the generative space to the conventional solutions the domain has already produced.

The evidence for this interpretation is in the mechanism data. Every non-artist condition in the isolation study (T1 through T6, A0, and implicitly the puzzle designer control referenced in the introduction) remained within the letter-extraction mechanism family. The artist condition alone escaped it. The escape was not produced by domain knowledge that the artist persona encoded—the artist persona encodes no puzzle knowledge—but by the experience-first orientation that asked "what does the solver do?" before asking "what letters spell the answer?" The answer to the first question was: reconstruction of damaged material, which is structurally isomorphic to the experiential act of deciphering something ancient. That answer could not be reached by starting with the conventions of puzzle hunt design, because those conventions do not include damaged-inscription reconstruction. It required designing from the solver's phenomenological position outward, rather than from the mechanism family inward.

### 5.2 A Hypothesis on Mechanism Family Escape

The artist condition was the only persona to escape the letter-extraction mechanism family (acrostic, indexing) that dominated all

other conditions. We propose the following hypothesis: experience-first orientation (T1) reframes the design problem before conventional mechanism templates activate. Under a domain label (“puzzle designer”), the model’s problem representation is “produce an AoE2 puzzle” — a representation that immediately activates the extraction family as the canonical puzzle hunt mechanism. Under an artistic orientation, the problem representation shifts to “produce an experience of discovery from damaged material” — a representation for which the letter-extraction family is not the obvious instantiation. This is analogous to Pérez y Pérez’s engagement/reflection phases in MEXICA [? ]: engagement (orientation-driven) determines which creative schemas are activated before reflection (evaluation) begins. T1 is the engagement operator; without it, generation proceeds directly to reflection over a default schema pool.

### 5.3 The Three-Trait Injection and Its Generalization

The minimum sufficient set (T1 + T4 + T5) has a practical form that can be stated in three sentences applicable to any creative brief across any domain: “Think first about how the other person will experience this. Remove anything that is not necessary. Verify every claim before finishing.”

These three sentences do not encode puzzle design knowledge, AoE2 unit costs, or letter-extraction conventions. They encode a way of working. The first sentence reorients from content delivery to experience design. The second reorients from accumulation to reduction. The third reorients from confident assertion to verified claim. Together, they constitute the operational core of quality-oriented creative work in any domain where a human being will eventually encounter the output.

This generalization has theoretical grounding in the literature the traits operationalize. Experience-first thinking (T1) is the application of Norman’s central design principle—that the user’s mental model is the primary constraint, not the structural properties of the artifact—to the creative generation context [? ]. Elegance drive (T4) is the application of Rams’s “as little design as possible” to the prompt-level instruction [? ]: the creator who removes anything that is not necessary is applying Rams’s discipline not to a physical product but to a generated artifact. Rigor and verification (T5) is the implementation hygiene that keeps the other two honest. Without T5, experience-first orientation can produce beautiful-but-broken puzzles (the solver’s experience is designed with care, but the clue that claims Watch Tower has a Blacksmith prerequisite is wrong, and the solver who knows this cannot solve the puzzle). Without T5, elegance drive can produce stripped-but-incorrect artifacts. The three traits are not independent quality contributions; they form a feedback loop. T1 sets the target (the solver’s experience), T4 enforces discipline toward that target (remove what does not serve it), and T5 verifies that what remains is true (claims that are false serve no target).

We predict that a business analyst applying these three sentences to a deliverable will produce a more audience-oriented, less cluttered, and more accurate artifact than one who does not, for the same structural reasons the three-trait puzzle passes the C8 panel and the two-trait puzzle does not (predicted; cross-domain

validation pending). The domain is irrelevant to this prediction; the cognitive stance is the operative variable.

### 5.4 Why Elegance Drive Backfired in Isolation

The T4 isolation result (26.0/45, below the null baseline of 24.0) is the study’s most counterintuitive data point, and it deserves a direct explanation. The elegance drive trait, applied alone, produced the study’s second-worst puzzle. The stripped-bracket presentation—five clues with the answer letters visible in brackets, no narrative, no misdirection—made the extraction mechanism so transparent that there was nothing for the solver to discover. The brackets presenting [T], [O], [W], [E], [R] in a vertical column give the solver the answer before a single clue is read. Stripping form without experience-first orientation strips the solver’s journey along with the clutter.

The elegance paradox this reveals is precise: T4 alone produces transparency, not clarity. Clarity is a property of the relationship between the solver’s understanding and the mechanism’s structure—the mechanism is clear when the solver’s journey to find it is unobstructed. Transparency is the elimination of that journey entirely. The difference between a clean puzzle and a transparent one is that the clean puzzle withholds until the solver earns the mechanism, while the transparent one removes the withholding along with the clutter.

This is why T4 needs T1 to know what to keep. Experience-first orientation identifies the things that serve the solver’s journey and the things that obstruct it; elegance drive then removes the obstruction. Without T1, T4 cannot distinguish journey from clutter and removes both. The three minimum sufficient traits are not additive; T4 operates as a constraint on an experience-designed space that T1 defines, and T5 operates as a verification pass on the contents of that space. The traits form a dependency chain, not an independent feature list.

### 5.5 Implications for AI Creative Pipeline Design

The findings suggest three practical recommendations for practitioners deploying AI creative generation systems.

First, replace domain labels with artistic orientation labels for creative tasks. “You are a puzzle designer,” “you are a marketing copywriter,” “you are a product specification author”—these labels activate domain conventions that constrain the generative space to what the domain has already produced. “You are an artist who works in this domain” or, more precisely, the three-sentence injection (T1, T4, T5)—activates the experience-design orientation that makes domain knowledge serve the audience rather than serve itself. The puzzle designer knows what an acrostic is; the artist asks whether an acrostic is what the solver needs.

Second, when domain expertise is genuinely required, inject it as content context rather than persona identity. The Phase 3 reconstruction experiments demonstrate this directly: the CALIBER puzzle (R3, T1+T4+T5) uses the game’s actual values as the mechanism not because the artist persona encoded AoE2 knowledge but because the world data file was available as a resource. Domain expertise as content context (factual reference material, world files, knowledge bases) allows the model to draw on domain knowledge without adopting the domain’s conventions as constraints on the

design stance. The model with T1+T4+T5 and access to a world file asks “how can this data serve the solver’s experience?” The model with a “puzzle designer” identity label asks “how can I build a puzzle using this data?” The first question produces CALIBER; the second produces The Indexed Garrison.

Third, the minimum three-sentence injection (T1+T4+T5) can be applied without domain-specific calibration. It does not reference puzzle design, AoE2, letter extraction, or any domain concept. It references only the cognitive stance toward quality that crosses the threshold in this study. We predict that practitioners who prepend these three sentences to any creative generation prompt—marketing brief, training material, interactive fiction specification, product design requirements—will observe a gap between domain-labeled and artistically-oriented output analogous to the 15.6-point finding documented here (predicted; cross-domain validation pending). The connection to Paper #3’s finding [?] is structural: that study demonstrates that authoring profile depth mirrors reviewing profile depth in its effect on output quality. The trait decomposition reported here explains why profile depth matters—richer profiles activate more of the six traits that constitute artistic quality orientation, and the traits that the richer profile activates more fully (T1, T4, T5) are precisely the ones whose absence causes the largest quality degradations.

## 6 Conclusion

This paper decomposed the “artist” persona advantage over the “puzzle designer” baseline into six constituent cognitive traits, tested each trait through three phases of controlled experimentation, and identified the minimum sufficient trait set for artist-level puzzle authoring quality. Three findings constitute the core result. First, experience-first thinking (T1) is the necessary but not sufficient condition for puzzle quality: its removal causes a 29.4-point categorical failure, producing not a degraded puzzle but an absent one—a list of AoE2 terms with no clue layer, no mechanism, and no solver task. Second, the minimum sufficient set is T1 + T4 + T5 (experience-first, elegance drive, rigor/verification), which crosses the 33/45 quality threshold at Phase 3 reconstruction step R3 (35.7/45). Tier 3 traits (visual thinking, surprise instinct, systematic thinking) each contribute 4–8 additional points above the minimum sufficient baseline but are not structurally irreplaceable. Third, domain labels activate template-matching; artistic orientation activates experience-design. The puzzle designer persona produced an acrostic with an extraction table because those are the conventions the label activates. The artist persona produced a damaged-inscription mechanism because it asked first what the solver would *do*, not what structure would yield the answer—and that reorientation is instantiated by the experience-first trait.

The practical application of this finding is a three-sentence creative quality injection that generalizes across domains: “Think first about how the other person will experience this. Remove anything that is not necessary. Verify every claim before finishing.” These sentences do not encode domain knowledge; they encode a way of working. The first reorients from content delivery to experience design, the second enforces reduction discipline, and the third provides implementation hygiene that keeps the other two honest. We predict that prepending these three sentences to any

creative generation prompt—marketing copy, product design specifications, legal plain-language drafting, training material authoring, interactive fiction, game level design—will replicate the threshold-crossing effect documented here. The artist knows not what to make but how to make it for someone; these three sentences inject that knowledge operationally, without reference to any creative domain’s conventions. The 15.6-point gap between domain-labeled and artistically-oriented AI output is not a puzzle-specific finding; it is a finding about the cognitive architecture of quality-oriented creative work, and the three-sentence injection is its operational form. These cross-domain predictions are grounded in the universal structure of creative receiver relationships but require empirical validation beyond puzzle hunt design before operational deployment.

Three directions merit future investigation. Cross-domain replication would test whether T1+T4+T5 produces the same threshold-crossing effect in business document generation, interactive fiction authoring, game level design, and other creative tasks where audience experience is the primary design constraint. A replication finding in two or more non-puzzle domains would establish the generalization claim on empirical rather than theoretical grounds. Multi-trait interaction analysis would test whether T1×T4 produces supralinear quality gains—the elegance paradox observed here (T4 below baseline in isolation, load-bearing in combination) suggests that trait dependencies may produce non-additive effects, and identifying which trait pairs interact positively or negatively would provide a more complete causal account of the artist advantage than additive decomposition alone provides. Human validation would ask whether human puzzle designers who explicitly apply the three-trait stance to their work self-report improved output quality, and whether the C8 panel’s quality judgments of human-authored puzzles align with the AI-authored findings reported here. The natural forward pointer is to the human-AI calibration question addressed in the companion paper [?]: whether T1+T4+T5 closes the residual gap between AI-authored and human-authored puzzle quality on the subjective dimensions (Fun, Elegance, Reading Reward) where the artist persona’s ceiling (42.3/45) still falls slightly short of the best human-authored conditions in that study is the question this work opens.

## References